MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 8590

This specification is approved for use by the Engineering and Technical Support Division (AFLC/LOIE), Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Tetrode, metal-ceramic
See figure 1
Mounting position: Any
Weight: 4 ounces nominal

ABSOLUTE MAXIMUM RATINGS: Fl = 500 MHz

Parameter: / Unit:	Ef V (note 1)	Eb Vdc	eb v	Ec1 Vdc	Ec2 Vdc	ec2 v	Ib mAdc	ib a (note 2)
Class C telegraphy:	6.0 ±5%	2,500	---	-250	500	---	250	----
Grid pulsed, Class C rf amp:	6.0 ±5%	5,500 (note 10)	---	-250	---	1,000	---	6.0
Anode pulsed, Class C rf amp:	6.0 ±5%	----	5,500	-250	---	1,000	---	6.0
Pulse mod or volt reg:	6.0 ±5%	7,000	---	-400	750	---	---	6.0
TEST CONDITIONS:	6.0	1,000	----	Adj	300	----	150	---

ABSOLUTE MAXIMUM RATINGS: Fl = 500 MHz

Parameter: / Unit:	tp µs (note 2)	Du ---- (note 2)	Pg1 W	Pg2 W	Pp W	T(anode core& seal) °C	tk sec (min)	Cooling --- (note 3)
Class C telegraphy:	---	----	2.0	12	250	250	30	---
Grid pulsed, Class C rf amp:	80	0.0016	2.0	12	250	250	30	---
Anode pulsed, Class C rf amp:	80	0.0016	2.0	12	250	250	30	---
Pulse mod or volt reg:	80	0.0016	2.0	12	250	250	30	---
TEST CONDITIONS:	---	----	---	---	---	---	120	Note 4

GENERAL:

Qualification - Required

8590

FSC 5960

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1261	Electrode voltage (grid)		0.65	II	Ec1	-35.0	-48.0	Vdc
1256	Electrode current (screen)		0.65	II	Ec2	-7.0	+3.0	mAdc
1266	Total grid current	Eb = 2,000 Vdc; Ec1/Ib = 125 mAdc	0.65	II	Ic1	---	-15	µAdc
1266	Primary grid emission (control)	Ic1 = 70 mAdc; t = 15; anode and g2 floating	0.65	II	Isg1	---	-25	µAdc
1266	Primary grid emission (screen)	Ec1 = 0; t = 15; Ic2 = 100 mAdc; anode floating	0.65	II	Isg2	---	-250	µAdc
1372	Current division (long pulse)	Eb = Ec2 = 250 Vdc; Ec1 = -100 Vdc; prr = 11 ±1; tp = 4,500 µs (min); egk/ib = 1.0 a	0.65	II	egk	5.0	16.0	v
					ic1	---	200	ma
					ic2	---	260	ma
	Quality conformance inspection, part 2							
1301	Heater current		---	---	If	2.3	3.0	A
1331	Direct-interelectrode capacitance (grounded grid connection)		---	---	Cpk	---	0.01	pF
			---	---	Cin	12.0	16.0	pF
			---	---	Cout	3.9	4.35	pF
[illegible]	Power gain	f = 144 ±5 MHz, ec2 = 1,000 v; Eb = 5,500 Vdc; Ec1 = -100 to -200 Vdc; Du = 0.005; tp = 250 µs; prr = 20; pd = 1,000 w (max); ib = 3.5 a (max) See note 6	[illegible]	[illegible]	po	[illegible]	[illegible]	kw useful
	Quality conformance inspection, part 3							
----	Life test	Group C; power gain except, pd= 1,000 w t = 500 hours See note 7	---	---	---	---	---	---
----	Life-test end points:							
----	Power gain		---	---	po	9.0	---	kw useful
			---	---	ib	---	3.5	a
1266	Primary grid emission (control)		---	---	Isg1	---	-100	µAdc
1266	Primary grid emission (screen)		---	---	Isg2	---	-250	µAdc
1155	Coolant-pressure drop versus coolant flow (forced air)	Eb = 1,000 Vdc; Ec1 = 0; Ec2/Ib = 250 mAdc; airflow = 4.8 cfm (max) See note 8 and 9	---	---	P	---	0.35	in.H_2O
					T(anode core)	---	250	°C

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 3 -Continued							
1031	Low-frequency vibration	No voltages See note 11	---	---	---	---	---	---
1042	Shock, specified pulse	No voltages; accel = 15G peak (min); D = 11 ±2 ms half-sine wave See note 11	---	---	---	---	---	---
----	Shock, and low-frequency vibration end points:							
1261	Electrode voltage (grid)		---	---	Ec1	-35.0	-48.0	Vdc
1266	Total grid current		---	---	Ic1	---	-15	µAdc
----	Resonance	No voltages applied See notes 5 and 8	---	---	F	465	489	MHz

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NOTES:

1. At frequencies above approximately 300 MHz under Class C telegraphy conditions, it may be necessary to reduce heater voltage to compensate for rf transmit-time heating of the cathode. This type of back-heating is a function of frequency, grid bias, anode current, duty cycle, and circuit design and adjustment. Heater voltage should be maintained within ±5 percent when long life and consistent performance are factors, but in no case shall the heater be operated at less than 5.4 volts. The following heater operation voltages are for straight-through CW amplifier operation:

Frequency (MHz)	Ef (V)
300 or lower	6.00
301 to 400	5.75
401 to 500	5.50

2. Peak anode current (ib) may be considered as average during the pulse and shall be limited to 6.0 amperes. With a pulse length (tp) longer than 80 µs or a duty ratio (Du) higher than 0.0016, peak anode current shall be reduced in accordance with the data shown on figure 2.

3. Forced-air cooling shall be provided for the base and anode. The maximum seal and anode core temperature rating should not be exceeded. At an anode dissapation of 250 watts and an incoming air temperature of 50°C maximum at sea level, a minimum airflow of 4.8 cfm shall pass through the anode cooler. With voltages applied to the tube, the static pressure drop across the anode cooler with an airflow of 4.8 cfm at 50°C at seal is approximately 0.25 inch of water. The pressure drop varies with the amount of escaping air, air temperature, and with the shape and construction of the air detector. The pressure drop in most equipments can be expected to be considerably higher than 0.25 inch of water because of temperature and system losses caused by associated hardware such as collets, cavity, etc. The stem shall be cooled by a portion of the anode air or by a separate air supply. Airflow should be applied before or simultaneously with electrode voltages (including the heater) and may be removed simultaneously with them. In cases where long life and consistent performance are factors, cooling in excess of minimum requirements is normally beneficial.

4. In all electrical tests involving application of heater voltage, forced-air cooling of the tube shall be allowable at the rate of 4.8 cfm maximum, for the base and anode, at sea level. Cooling air shall not have a temperature less than 20°C.

5. Test cavity per Drawing No. 2019229-0501 (The Bendix Corporation, Communications Division), or equivalent. With the tube inserted in the cavity, a rf signal from a standard signal generator is coupled into the cavity by means of a small loop. Signal is extracted from the cavity by means of another small loop, detected, and displayed on an HP-415-B, or equivalent, standing-wave ratio indicator. The signal generator frequency is adjusted for maximum output after detection. The frequency is then measured by means of a good quality wavemeter, such as the PRD Electronics Inc. Model 1587A, or equivalent.

6. Test cavity per Drawing No. 2059524-0501 or No. 2019230-0501(The Bendix Corporation, Communications Division), or equivalent. Cooling air need not be restricted to 4.8 cfm during this test because of system losses. Cavity tuned so as to have a 3 dB bandwidth of 25 MHz centered at 442 MHz.

7. When used in the AN/FPS-85, this tube shall have a 4,000-hour service-life guarantee. The life of the tube shall be considered terminated when one of the following conditions exist:

   (a) There is an accumulation of 10-anode arcs.

   (b) The power output has dropped to the 3 dB level.

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NOTES: -Continued

8. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test shall be made as a part of Quality Conformance Inspection, Part 2, Code Level D, with an AQL of 6.5. The regular "12-calendar months" double sampling plan shall be reinstated after three consecutive samples have been accepted.

9. Pressure drop shall be measured across the anode cooler only, with the tube operating. Anode core temperature shall be measured with the tube operating, with an ambient air temperature of 25°C, with both base and anode cooled by applying an airflow of 4.8 cfm maximum at sea level. Temperature shall be measured by means of a thermocouple attached to the top of the anode core, adjacent to the cooler, by any suitable means. Good electrical continuity between the thermocouple and the metal area in proximity shall be demonstrated before the cooling test is performed.

10. Eb of 6,300 Vdc (maximum) is permissible under zero drive conditions.

11. Test to be performed every three months, using the following sampling plan:

n1 = 4	c1 = 0	Where c2 represents the total allowable failures
n2 = 4	c2 = 1	for the first and second samples combined.

Separate samples may be used at the option of the manufacturer, and the tests shall be considered as nondestructive except in case of failure. In case of failure, after double sampling, the failing test shall become a Quality Conformance Inspection, Part 2, AQL 6.5, Inspection Level S3, for three consecutive successful submissions, at which time the testing may revert to the once every three months testing basis.

12. Reclaimed materials shall be utilized to the maximum extent possible.

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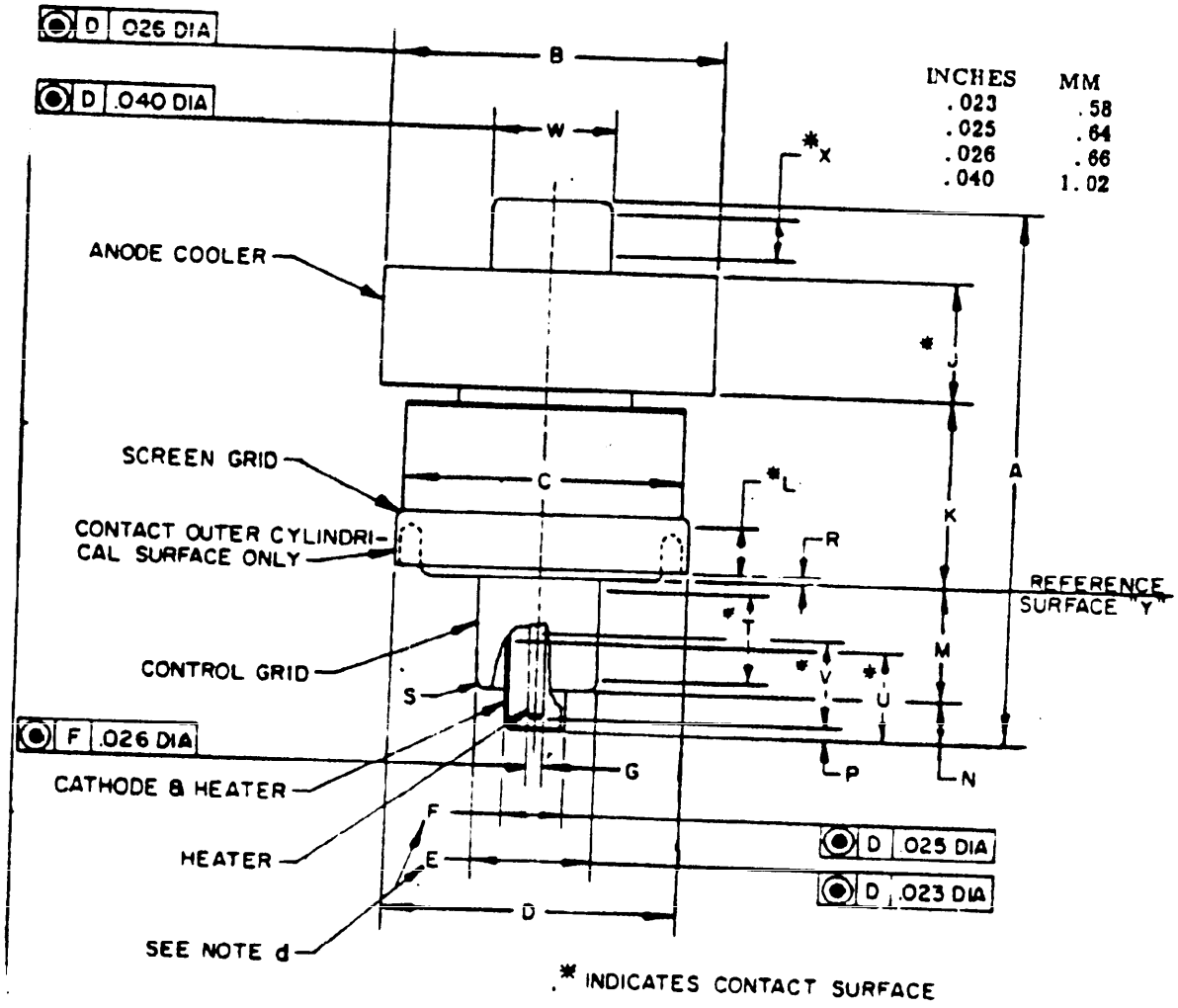
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NOTES:

a. The eccentricity values shown shall be considered as maximum values, and shall be checked on a Quality Conformance Inspection, Part 3 (basis), with the provisions of note e applicable.

b. The tube shall be rotated on diameter D when eccentricity is being measured.

c. Surface Y shall be perpendicular to the measuring platform when eccentricity is being measured.

d. Average diameter of E shall be as noted, and may be out of round a total of .006 (.15 mm). Average diameter of F shall be as noted, and may be out of round .006 (.15 mm).

e. Dimensions listed under Quality Conformance Inspection, Part 3 shall be checked twice yearly, using a sample of 10 tubes with 1 failure allowed. In case of a sample failure, the failing dimension(s) shall become Quality Conformance Inspection, Part 2, AQL 6.5, Inspection Level S3, for three consecutive successful submissions, at which time the test may revert to the periodic-check test basis.

FIGURE 1. Outline drawing of electron tube type 8590.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A		2.813 (71.45)
D	1.415 (35.94)	1.435 (36.45)
Quality conformance inspection, part 3 (periodic-check) (see note e)		
B	1.615 (41.02)	1.640 (41.66)
C		1.406 (35.71)
E	.588 (14.94)	.597 (15.16)
F	.318 (8.08)	.325 (8.26)
G	.091 (2.31)	.095 (2.41)
J	.585 (14.86)	.665 (16.89)
K	.900 (22.86)	.950 (24.13)
L	.187 (4.75)	
M	.520 (13.21)	.560 (14.22)
N	.235 (5.97)	.265 (6.73)
P	.032 (.81)	.082 (2.08)
R		.040 (1.01)
S		.171 (4.35) Rad
T	.388 (9.86)	
U	.406 (10.31)	
V	.468 (11.89)	
W	.559 (14.20)	.573 (14.55)
X	.240 (6.10)	

FIGURE 1. Outline drawing of electron tube type 8590 - Continued

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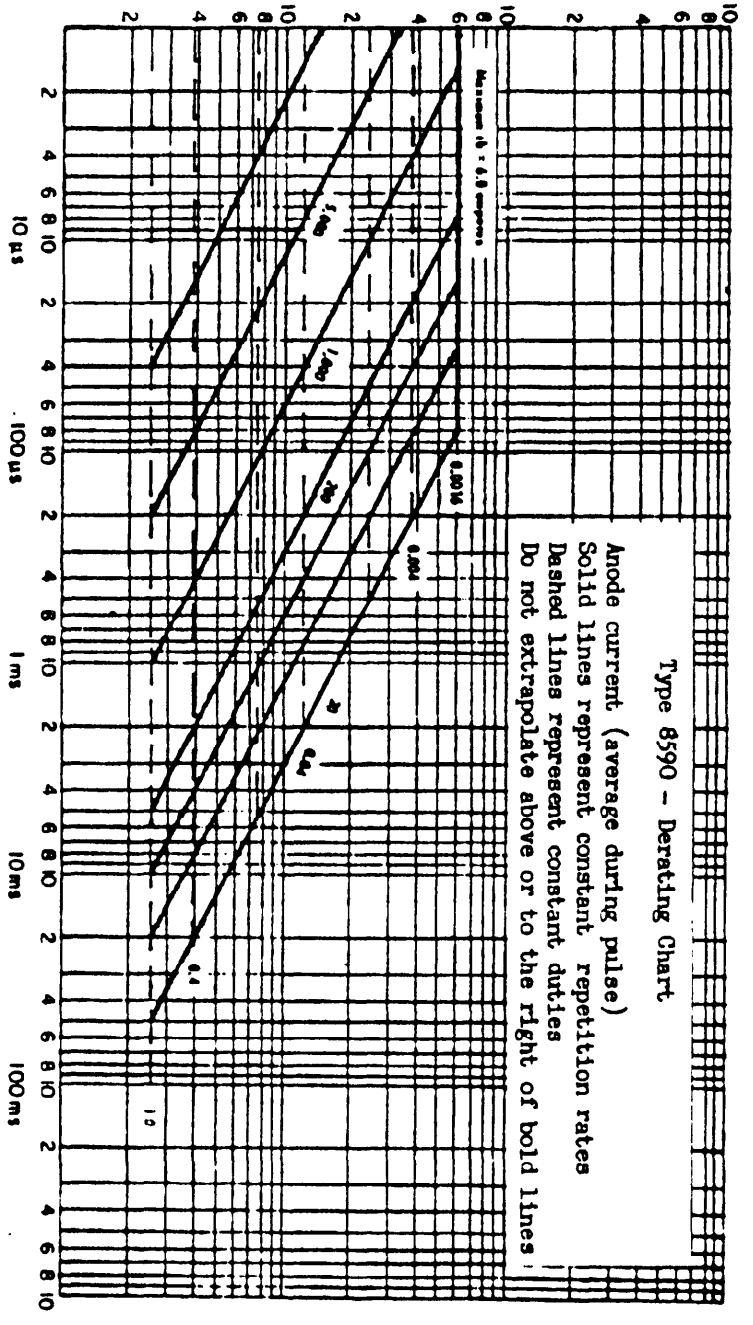


FIGURE 2. Derating chart.

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